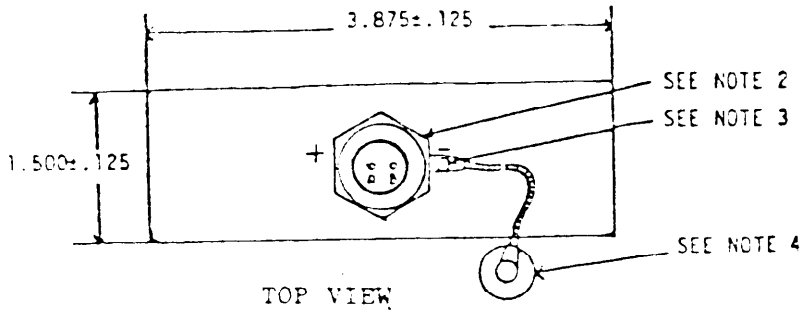
MILITARY SPECIFICATION SHEET

BATTERY, RECHARGEABLE, SEALED NICKEL CADMIUM BB-541/U

This specification sheet is approved for use by the Electronic Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sealed nickel cadmium rechargeable battery type described herein shall consist of this document and the latest issue in effect of Specification MIL-B-49436(ER).



INCHES	MM
.125	3.18
1.500	38.10
3.875	98.43
7.000	177.80
8.250	209.55

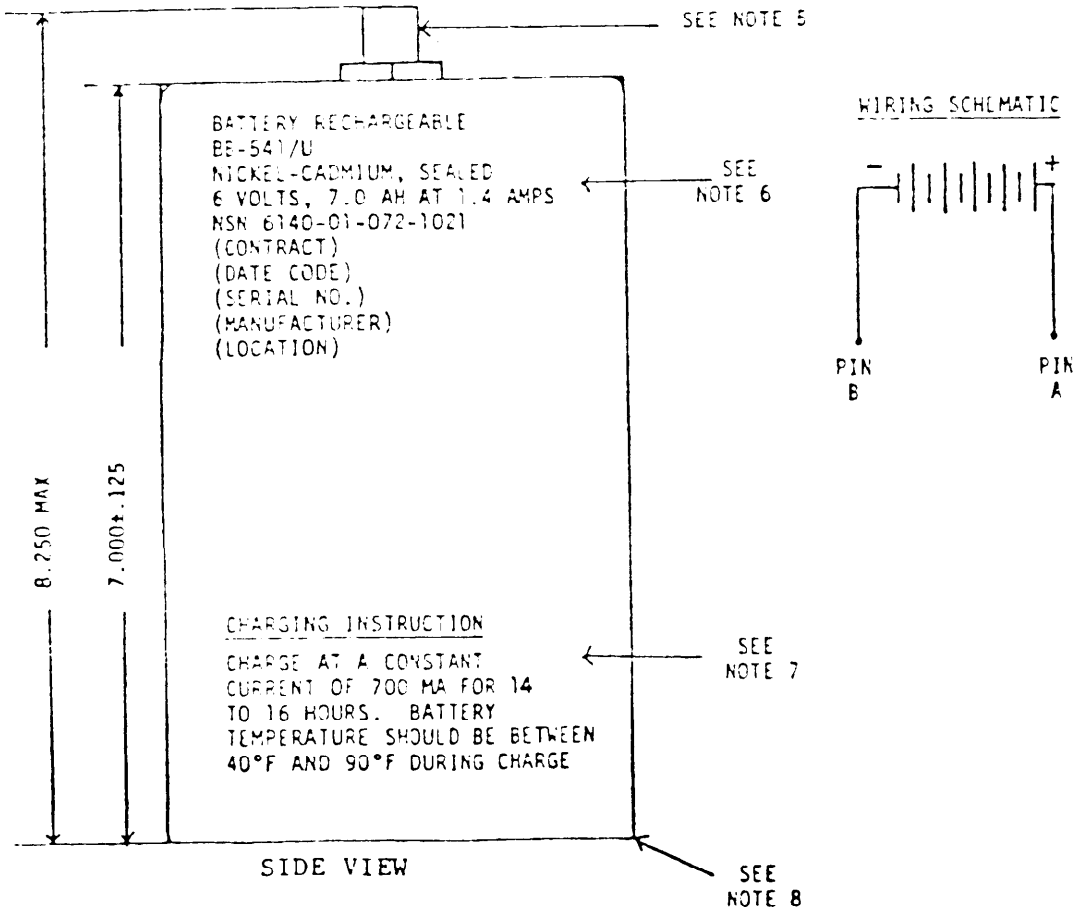


FIGURE 1. DIMENSIONS AND LABELS FOR BB-541/U

FSC-6140

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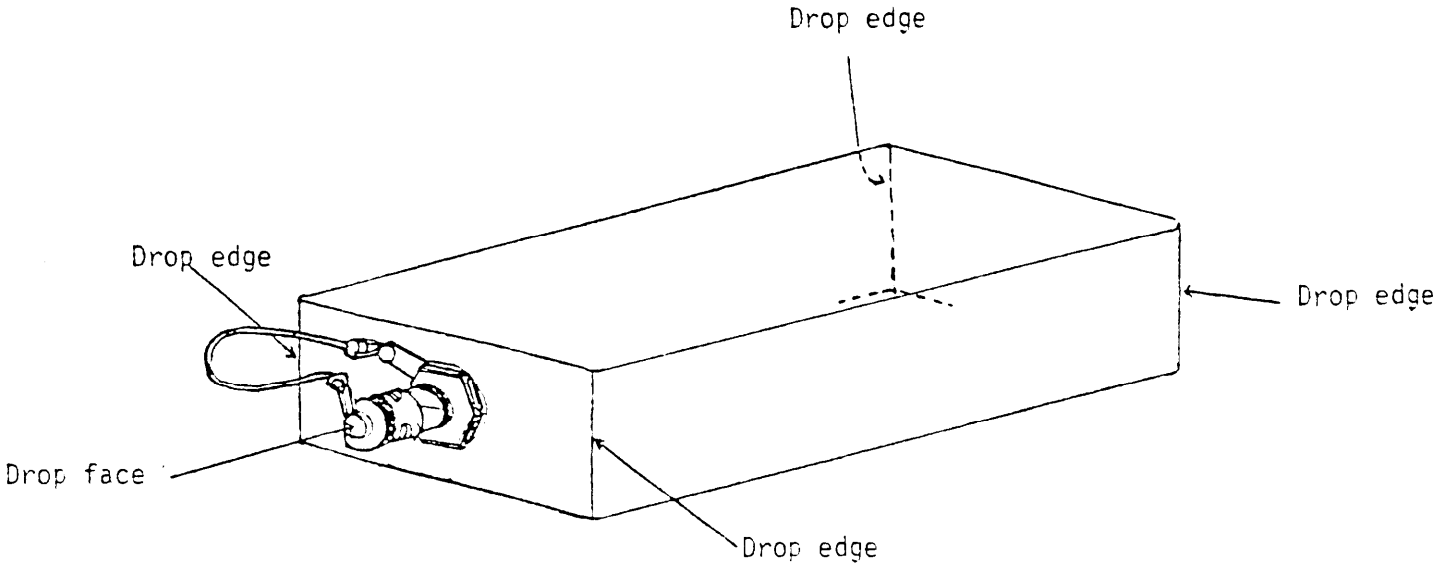


FIGURE 2. Drop sites for drop test

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DRAWING NOTES

1. Dimensions in inches.
2. Jam nut, torque to 50 ± 5 in-lb.
3. Rivet.
4. Protective cap, BENDIX Part No. 10-215964-83, or equal.
5. Connector, BENDIX Part No. PT07C-8-2S, or equal.
6. Stencil (or equivalent) identification as shown. Manufacturer must fill in applicable information in parenthesis. Marking color shall be white.
7. Stencil (or equivalent) charging instructions in the location shown. Color marking shall be white.
8. Rounded corners.

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REQUIREMENTS:

Rated capacity (C_5):	7.0 AH
Voltage (Nominal):	6.0 Volts
Weight (Maximum):	1.36 Kg (3.0 lbs)
Connector:	BENDIX PT07C-8-2S receptacle or equal
Number and type of cells:	5 sealed "F" size cells or equal in series
Charge rate for testing and on battery (C/10 for 14 hrs):	0.7 amps

Discharge Rates	5 Second Voltage(V)	Final Voltage(V)	Current (Amps)
C/5	-	5.0	1.4
0.4 C_5	-	5.6	2.6
C_5	5.75	5.0	7.0

Test	Discharge Rate	Capacity Requirement(AH)
Full Capacity Discharge	C/5	7.0
Cycle Life, 28 cycles	C/5	6.3
Overcharge, 24 hours	C/5	7.0
High Rate Discharge	C_5	6.0
Low Temperature Discharge at -18°C (0°F)	.4 C_5	4.2
Retention of Charge 7 days at 51°C (124°F)	C/5	2.8
Pulse Discharge	Not applicable	-
Vibration	C/5	7.0

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REQUIREMENTS:

Mounting screws for connector. Not applicable.

Pressure release device. Not applicable.

Connector. Each battery shall be provided with a special purpose receptacle, BENDIX Part No. PT07C-8-2S or equal.

Receptacle mounting. Jam nut mounting shall be employed.

Receptacle, construction. Receptacle shall be of aluminum, cadmium plated in accordance with QQ-P-416, then chromate after treatment.

Cap, protection. Each battery shall be provided with a permenantly connected protective cap, BENDIX Part No. 10-215964-83, or equal, to mate with receptacle. The protection cap shall be attached by means of a retainer cable.

Color. The color of the battery shall be black.

Battery case. In accordance with MIL-B-49436 except that the case shall be ABS plastic conforming to MIL-B-19904, grade FR haircell, grains or equal. Minimum case thickness shall be .06 inches.

TESTS:

Cycle life test. In accordance with MIL-B-49436 except that:

a. Cycle 1- Charge at 0.7 amps for 14 hours, discharge at 2.6 amps for 2 hours.

b. Cycle 2 thru 26- Charge battery for 10 hours at 0.7 amperes and immediately discharge at 2.6 amperes for 2 hours.

c. Cycle 27- Charge battery for 10 hours at 0.7 amperes and immediately discharge at 1.4 amperes to 5.0 volts OCV.

d. Cycle 28- Charge battery at 0.7 amps for 14 hours, rest 2 hours and discharged at 1.4 amperes to 5.0 volts OCV.

e. The test shall be terminated when one of the following occurs:

(1) When battery is discharged at 2.6 amps for 2 hours, battery is less than 6 volts during discharge.

(2) The capacity of the 28th cycle is less than 6.3 ampere hours.

(3) The battery fails to pass electrical leakage.

(4) Twenty eight satisfactory cycles are completed.

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Helium leak test. The battery shall be tested by the following method:

a. Equipment-helium tank, pressure vessel (capable of 20 psig min), regulator and immersion vessel (water containment).

b. Place battery into pressure vessel and seal access part of pressure vessel.

c. Increase pressure in pressure vessel to $10 \pm {}^{1}_{0}$ psig of gaseous helium. Record pressure and time.

d. Expose the battery to a pressure of $10 \pm {}^{1}_{0}$ psig of gaseous helium for two hours. Release pressure in pressure vessel and record time.

e. Open access port and immediately transfer the battery into immersion vessel. Insure that battery is placed so that it is visible, the seam is facing up, and that the battery is covered by 1 ± 1/4 inch of water.

f. The battery shall be carefully observed (particularly at the seals) for a total of five minutes of the immersion for indication of leakage as evidenced by a steady stream of bubbles emanating from the specimen. Disregard surface adhering bubbles which do not change in size. Any indication of leakage shall be cause for rejection.

g. Remove battery from the immersion vessel, dry with a soft absorbent cloth and place on a table to air dry. Conduct full charge capacity test and meet the environmental requirements.

Drop test. A drop test shall be conducted in addition with the mechanical shock test. A charged battery shall be suspended so that one edge or terminal is 18 inches above a concrete surface and allowed to fall freely. The test shall be conducted on one face and four edges (see figure 3) per sample battery. Following the test, the batteries shall meet the requirements of 3.12.

Immersion test. Replaced by Helium leak test.

CUSTODIAN:
ARMY-ER
REVIEWER:
ARMY - CR, AV, GL

PREPARING ACTIVITY:
ARMY-ER
(PROJECT 6140-A577-5)

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